

INTRODUCTION

Problem Statement

Traditional safe haven assets, such as gold and silver, have historically been trusted as shields against economic uncertainty. Yet recent market shocks have exposed significant volatility in these assets, undermining investor confidence in their ability to preserve capital. As markets grow more complex and interconnected, there is a pressing need to identify modern alternatives that can reliably serve as stores of value during periods of systemic stress.

Research Question

What assets serve as superior stores of value to gold in the modern economy, offering capital preservation during crisis periods while delivering inflation beating returns over a five-year horizon?



Gold Closing Price (January 2021 – March 2026)

METHODS

To evaluate each asset as a potential safe haven, we selected five metrics that together capture the dimensions most critical to capital preservation: return quality, downside risk, inflation resilience, crisis behavior, and independence from the broader market. Rather than relying on any single measure, each asset was scored across all five metrics using percentile-based normalization, where scores reflect each asset's standing relative to the full pool of assets analyzed.

Assets at or above the 90th percentile in a given metric received a score of 10, with all others scaled proportionally to their percentile rank.

Assets Analyzed

Category	Tickers
Broad Market ETFs	SPY, VTI
Sector ETFs	XLU
Individual Stocks	NVDA, AAPL, MSFT, AMD, AMZN, TSLA, WMT, LOW, HD, JNJ
Crypto ETFs	BITW, IBIT, ETHA
Metals ETFs	GLD, SLV, PPLT, PALL
Agriculture ETFs	WEAT, SOYB, DBA

Real Returns (Inflation-Adjusted Returns)

This metric adjusts each asset's returns against CPI data to reflect what investors actually gained in real terms. An asset that outpaces inflation preserves wealth; one that doesn't is effectively losing value even if its price is rising.

Risk-Adjusted Return (Sortino Ratio)

The Sortino Ratio measures how much return an asset generates relative to the downside risk it carries. Unlike simpler measures, it only penalizes volatility that results in losses, not gains. A higher Sortino Ratio indicates an asset is delivering strong returns without exposing investors to excessive risk of losses.

Drawdown Protection (Calmar Ratio)

The Calmar Ratio compares an asset's annualized return against its worst peak-to-trough decline over the study period. A higher ratio means the asset achieved meaningful gains without suffering catastrophic drops, rewarding assets that protect capital even when the market is negative.

Crisis Performance (Breakeven Return Rate / Time to Recovery)

This metric evaluates how quickly an asset recovered from its steepest losses during documented market stress events. Assets with a low Breakeven Return Rate bounced back faster and more efficiently, signaling genuine resilience rather than just temporary stability.

Market Independence (Pearson Correlation vs. VTI)

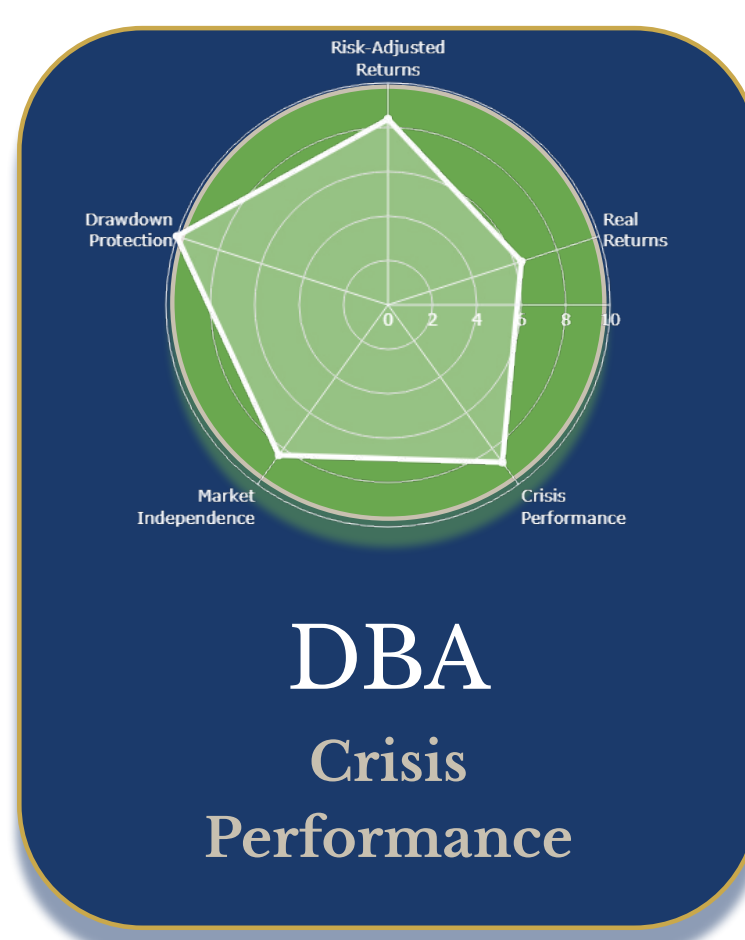
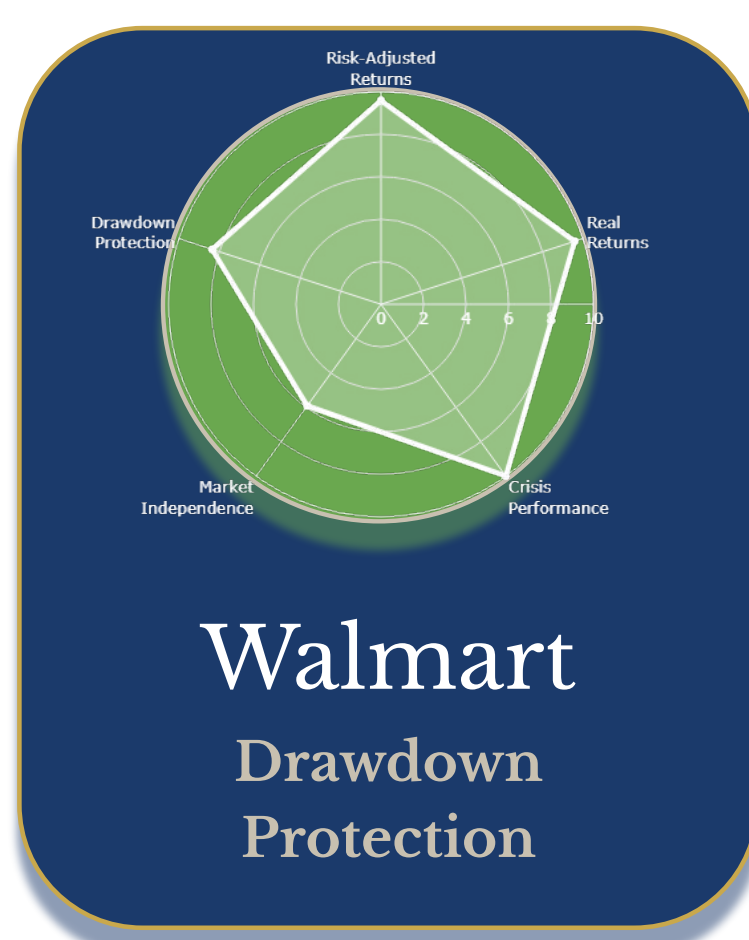
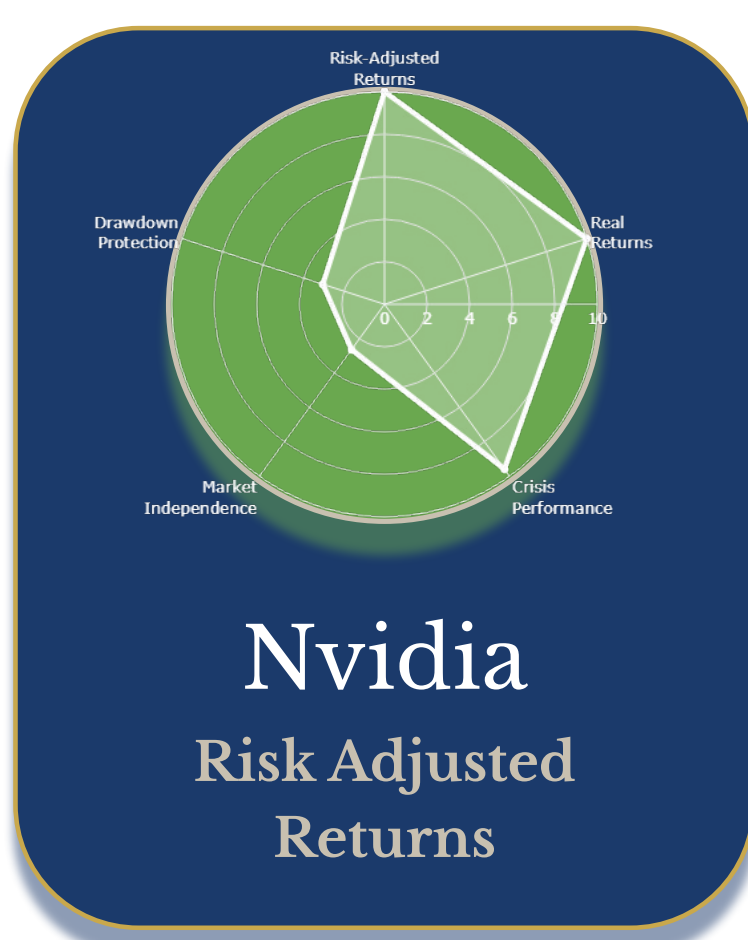
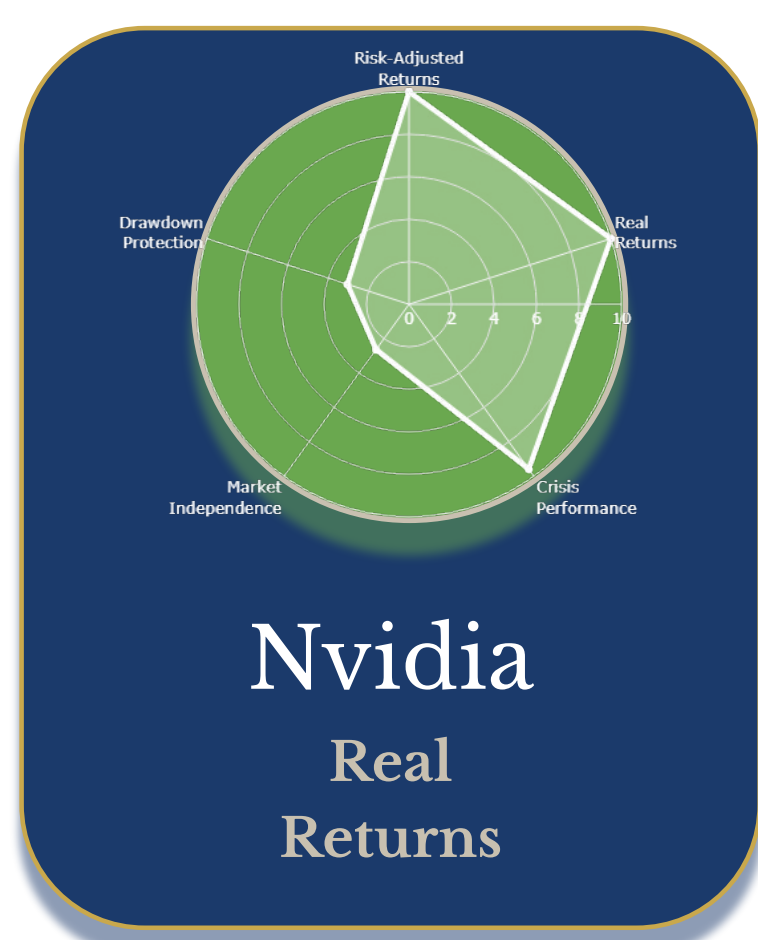
This metric measures how closely an asset's price movements track the total U.S. stock market, using the Vanguard Total Market ETF (VTI) as a benchmark. Assets with low or negative correlation tend to hold their value — or even rise — when the broader market falls, making them stronger candidates for true diversification.

Rank	Ticker	Category	Overall_Score
1	GLD	Metals	9.02
2	WMT	Stocks	8.69
3	DBA	Agriculture	8.36
4	SLV	Metals	7.47
5	NVDA	Stocks	7.06
6	XLU	Sector	6.97
7	IBIT	Crypto	6.81
8	PPLT	Metals	6.64
9	AAPL	Stocks	6.15
10	JNJ	Stocks	5.99
11	AMD	Stocks	5.83
12	VTI	Market	5.81
13	SPY	Market	5.75
14	SOYB	Agriculture	5.26
15	MSFT	Stocks	4.76
16	WEAT	Agriculture	3.95
17	LOW	Stocks	3.78
18	TSLA	Stocks	3.78
19	PALL	Metals	3.54
20	BITW	Crypto	3.37
21	AMZN	Stocks	2.96
22	ETHA	Crypto	2.80
23	HD	Stocks	2.72

BEST OVERALL



BEST IN CATEGORY



We acknowledge the assistance of Dr. Bill Eberle and Mr. Casey Martens

RESULTS & DISCUSSION

Results

Applying our five-metric percentile scoring framework to the 23-asset universe yielded a definitive ranking for modern stores of value. Despite concerns regarding its recent volatility, Gold (GLD) retained its position as the premier safe haven, achieving the highest overall score of 9.02 out of 10.

However, modern alternatives emerged as highly competitive substitutes. Walmart (WMT) secured the second rank with an overall score of 8.69, and the Invesco DB Agriculture Fund (DBA) ranked third at 8.36.

A metric-by-metric breakdown reveals diverse category champions. Nvidia dominated both risk return adjusted and real returns by delivering massive, volatility-adjusted upside. Walmart excelled in drawdown protection, effectively protecting capital during severe market drops. For diversification, the Teucrium Wheat Fund demonstrated the strongest market independence. Finally, the Invesco DB Agriculture Fund led in crisis performance, recovering from systemic shocks faster than traditional safe havens.

Discussion

While Gold remains the benchmark for capital preservation, our framework proves that non-traditional assets can serve as highly viable safe havens. Defensive consumer equities (WMT) and agricultural commodities (DBA) offered robust drawdown protection and market independence rivaling precious metals.

Conversely, the results exposed the limitations of emerging digital assets. Despite being labeled “digital gold,” crypto ETFs and cyclical tech stocks behaved as volatile, risk-on assets during crises. Ultimately, investors seeking safe-haven assets with gold-like resilience should consider defensive consumer equities and agricultural commodities.

SOURCES

- Stooq - Historical market pricing database
- FRED - Customer Price Index
- Yahoo Finance - yfinance library